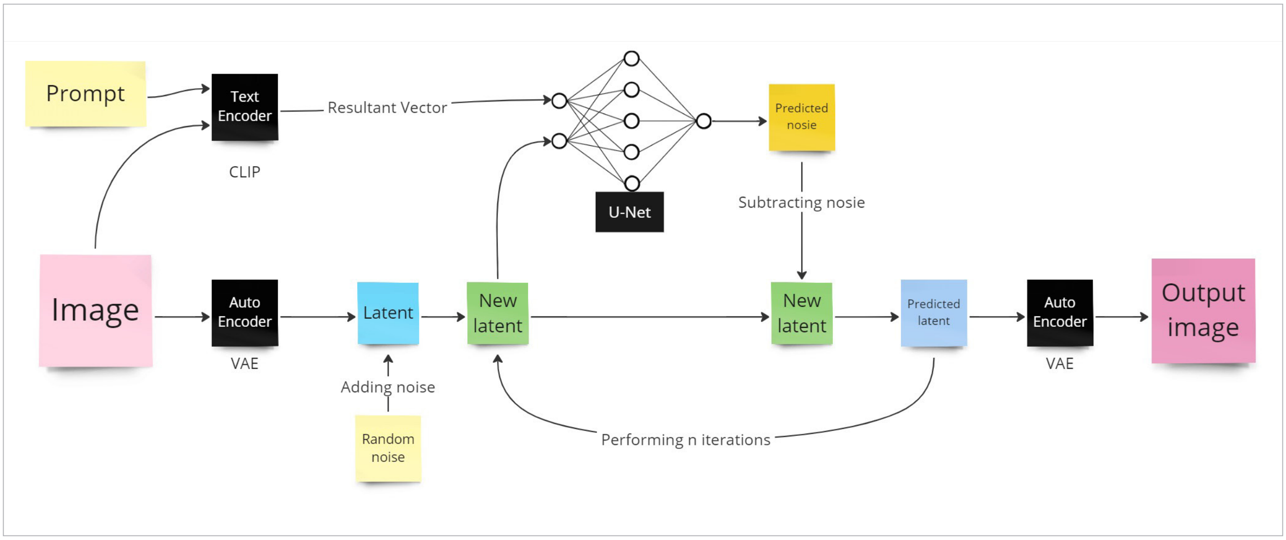

Figure 1: Flowchart of a Stable Diffusion model

Applications of Stable Diffusion

Stable Diffusion, an open source generative AI model, generates images based on text. It is built on the latent diffusion model by engineers and researchers from CompVis, LMU and RunwayML. Training in the early stages used 512x512 images from the LAION-5B database. Some of its applications include:

Image inpainting: The technique of repairing or replacing damaged areas in a picture is called inpainting. Images can be painted while keeping their general composition and substance intact by using Stable Diffusion.

Generative modelling: Stable Diffusion models can be applied for generative tasks like creating fresh, realistic samples from a given distribution. Tools like image synthesis can benefit from this.

Texture synthesis: In computer graphics and game development, Stable Diffusion may be used to synthesise textures which are detailed and look real.

Handling uncertainty: Stable Diffusion offers a probabilistic framework for managing uncertainty in the inpainting process. Compared to deterministic approaches, it offers a more robust approach and represents variations in the pixel values.

Noise reduction: Noise in images may take several forms, including random pixel changes and sensor noise. Stable Diffusion efficiently reduces noise while keeping crucial features.

How Stable Diffusion works

The principle of Stable Diffusion is to remove noise from a given image according to the prompt. A prompt is a text description of the operation to be performed. At Stable Diffusion's core are three main models: VAE (variational autoencoder), text-encoder (CLIP's text encoder) and U-net.

Text-encoder (CLIP's text encoder): CLIP stands for contrastive language-image pretraining. It is trained on image-text pairs. Image and text are encoded into two different vectors. CLIP returns the dot product of these two vectors. The resulting vector signifies the relation between image and text. This vector serves as an input to U-net.

U-net: Originally designed for image segmentation, in Stable Diffusion, U-net predicts the image noise based on input vectors.

The main issue here is the size of the image. Suppose you have a 512x512 image with three colour channels — red, blue and green; then there will be 512*512*3 inputs for the image to the U-net, which requires a significantly large number of resources. The solution to this problem is VAE.

VAE (variational autoencoder): VAE converts image representation into a low-dimensional representation called latent and vice-versa. In the above example, VAE converts the 512x512x3 image into 64x64x4 latent.

As shown in Figure 1, you are first required to input an image and prompt. Next, the text encoder encodes the prompt and the images into a vector. Simultaneously, the image is converted to latent by VAE. Now, random noise is added in latent. The newly formed latent is fed to the U-net along with the vector. U-net predicts the noise. The prediction is subtracted from the newly formed latent. The resultant latent is again fed to U-net with the same vector for 'n' times, at least 20 times. In every iteration, some noise is removed. After these iterations, the final latent is converted to image by VAE. This image is your final generated image.

Employing image masking

Image masking is a method used in photo editing to separate or isolate